

COMSATS University Islamabad

Sahiwal Campus

(Department of Computer Science)

Artificial Intelligence

Lab Assignment # 1

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Lab 01

Activity 1:

Let us take an integer from user as input and check whether the given value is even or not.

```
n = input ("Enter a number")
if int(n)%2==0:
  print("The given number is an even number")
else:
  print("The given number is an odd number")
```

```
Enter a number56
The given number is an even number
```

Activity 2:

Write a Python code to keep accepting integer values from user until 0 is entered. Display sum of the given values.

```
sum = 0
s = input("Enter an integer value...")
n = int(s)
while n!=0:
  sum=sum+n
  s = input("Enter an integer value...")
  n = int(s)
print("The sum of the given values is",sum)
```

```
Enter an integer value...1
Enter an integer value...2
Enter an integer value...0
The sum of the given values is 3
```

Activity 3:

Write a Python code to accept an integer value from user and check that whether the given value is prime number or not.

```
isPrime = True
i = 2
n = int(input("enter a number"))
while i < n:
    remainder = n % i
    if remainder == 0:
        isPrime = False
        break
    else:
        i = i + 1

if isPrime:
    print("Number is Prime")
else:
    print("Number is not Prime")
```

```
enter a number11
Number is Prime
```

Activity 4:

Accept 5 integer values from user and display their sum. Draw flowchart before coding in python.

```
summ = 0
i = 0
while i <= 4:
    s = input("enter a number")
    n = int(s)
    summ = summ + n
    i = i + 1

print("sum is ", summ)
```

```
enter a number13
enter a number1
enter a number4
enter a number11
enter a number12
sum is  41
```

Activity 5:

Calculate the sum of all the values between 0-10 using while loop.

```
summation = 0
i = 1
while i <= 10:
    summation = summation + i
    i = i + 1

print("sum is ", summation)
```

```
sum is  55
```

Activity 6:

Take input from the keyboard and use it in your program.

```
name = input('What is your name? ')
print('Hello ' + name)

job = input('What is your job? ')
print('Your job is ' + job)

num = input('Give me a number? ')
print('You said: ' + str(num))
```

```
What is your name? Hamza
Hello Hamza
What is your job? Student
Your job is Student
Give me a number? 7
You said: 7
```

Activity 7:

Generate a random number between 1 and 9 (including 1 and 9). Ask the user to guess the number, then tell them whether they guessed too low, too high, or exactly right.

```python
import random

MINIMUM = 1
MAXIMUM = 9
NUMBER = random.randint(MINIMUM, MAXIMUM)
GUESS = None
ANOTHER = None
TRY = 0
RUNNING = True

print("Alright...")

while RUNNING:
    GUESS = input("What is your lucky number? ")
    if GUESS.lower() == "exit":
        print("Better luck next time.")
        RUNNING = False
    else:
        GUESS = int(GUESS)
        if GUESS < NUMBER:
            print("Wrong, too low.")
        elif GUESS > NUMBER:
            print("Wrong, too high.")
        elif GUESS == NUMBER:
            print("Yes, that's the one, %s." % str(NUMBER))
            if TRY < 2:
                print("Impressive, only %s tries." % str(TRY))
            elif TRY > 2 and TRY < 10:
                print("Pretty good, %s tries." % str(TRY))
            else:
                print("Bad, %s tries." % str(TRY))
            RUNNING = False
    TRY += 1
```

```
Alright...
What is your lucky number? 9
Wrong, too high.
What is your lucky number? 8
Wrong, too high.
What is your lucky number? 7
Yes, that's the one, 7.
Bad, 2 tries.
```

Lab Task 1:

Write a program that prompts the user to input an integer and then outputs the number with the digits reversed. For example, if the input is 12345, the output should be 54321.

```
arr = [1, 2, 3, 4, 5]

for i in range(5):
    print(arr[i], end=" ")

print()

for i in range(4, -1, -1):
    print(arr[i], end=" ")
```

```
1 2 3 4 5
5 4 3 2 1
```

Lab Task 2:

Write a program that reads a set of integers, and then prints the sum of the even and odd integers.

```python
even_sum = 0
odd_sum = 0

count = int(input("How many numbers will you enter? "))

for i in range(count):
    num = int(input("Enter number: "))

    if num % 2 == 0:
        even_sum += num
    else:
        odd_sum += num

print("Sum of even numbers:", even_sum)
print("Sum of odd numbers:", odd_sum)
```

```
How many numbers will you enter? 2
Enter number: 2
Enter number: 7
Sum of even numbers: 2
Sum of odd numbers: 7
```

Lab Task 3: Fibonacci series is that when you add the previous two numbers the next number is formed. You have to start from 0 and 1.

```python
n = int(input("How many terms of Fibonacci series do you want? "))

a, b = 0, 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
print()
```

```
How many terms of Fibonacci series do you want? 2
0 1
```

Lab Task 4:

Write a Python code to accept marks of a student from 1-100 and display the grade according to the following formula.

```
marks = int(input("Enter marks (1-100): "))

if marks < 50:
    grade = "F"
elif 50 <= marks <= 60:
    grade = "E"
elif 61 <= marks <= 70:
    grade = "D"
elif 71 <= marks <= 80:
    grade = "C"
elif 81 <= marks <= 90:
    grade = "B"
elif 91 <= marks <= 100:
    grade = "A"
else:
    grade = "Invalid marks"

print("Grade:", grade)
```

```
Enter marks (1-100): 50
Grade: E
```

Lab Task 5:

Write a program that takes a number from user and calculate the factorial of that number.

```
num = int(input("Enter a number to find its factorial: "))

factorial = 1

for i in range(1, num + 1):
    factorial *= i

print("Factorial of", num, "is", factorial)
```

```
Enter a number to find its factorial: 4
Factorial of 4 is 24
```

Lab 02

Activity 1

Accept two lists from user and display their join.

```
myList1 = []
print("Enter objects of first list...")
for i in range(5):
    val = input("Enter a value: ")
    n = int(val)
    myList1.append(n)

myList2 = []
print("Enter objects of second list...")
for i in range(5):
    val = input("Enter a value: ")
    n = int(val)
    myList2.append(n)

list3 = myList1 + myList2
print(list3)
```

```
Enter objects of first list...
Enter a value: 1
Enter a value: 1
Enter a value: 1
Enter a value: 1
Enter a value: 1
Enter objects of second list...
Enter a value: 1
Enter a value: 1
Enter a value: 1
Enter a value: 1
Enter a value: 1
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
```

Activity 2:

A palindrome is a string which is same read forward or backwards.

```
def isPalindrome(word):
    temp = word[::-1]
    if temp.capitalize() == word.capitalize():
        return True
    else:
        return False

print(isPalindrome("deed"))
```

```
True
```

Activity 3:

Write a python code that finds another matrix/2D list that is a product of and b, i.e., C=a*b

```
a = [[1, 0, 0], [0, 1, 0], [0, 0, 1]]
b = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]

c = []
for indrow in range(3):
    c.append([])
    for indcol in range(3):
        c[indrow].append(0)
        for indaux in range(3):
            c[indrow][indcol] += a[indrow][indaux] * b[indcol][indaux]

print(c)
```

```
[[1, 4, 7], [2, 5, 8], [3, 6, 9]]
```

Activity 4:

Write a python function that takes a list of N tuples as input and returns the perimeter of the polygon. Remember that your code should work for any value of N.

```
def perimeter(listing):
    leng = len(listing)
    perimeter = 0
    for i in range(0, leng-1):
        dist = (((listing[i][0]-listing[i+1][0])**2) + ((listing[i][1]-listing[i+1][1])**2))**0.5
        perimeter = perimeter + dist
    perimeter = perimeter + (((listing[0][0]-listing[leng-1][0])**2) + ((listing[0][1]-listing[leng-1][1])**2))**0.5
    return perimeter

L = [(1,3), (2,7), (4,9), (-1,8)]
print(perimeter(L))
```

```
17.43571707109114
```

Activity 5:

Imagine two sets A and B containing numbers. Without using built-in…

```
def symDiff(a, b):
    c = set()
    for i in a:
        if i not in b:
            c.add(i)
    for i in b:
        if i not in a:
            c.add(i)
    return c

set1 = {0, 1, 2, 4, 5}
set2 = {4, 5, 7, 8, 9}
print(symDiff(set1, set2))

print(set1.symmetric_difference(set2))
print(set2.symmetric_difference(set1))
print(set1 ^ set2)
print(set2 ^ set1)
```

```
{0, 1, 2, 7, 8, 9}
{0, 1, 2, 7, 8, 9}
{0, 1, 2, 7, 8, 9}
{0, 1, 2, 7, 8, 9}
{0, 1, 2, 7, 8, 9}
```

Activity 6:

Create a Python program that contains a dictionary of names and phone numbers.

```
sample = {("sohaib", "ali"): "0246585468445", ("aib", "li"): "024658546645", ("sib", "ai"): "0246585468454"}
firstName = input("enter first name")
lastName = input("enter last name")

searchTuple = (firstName, lastName)
if searchTuple in sample:
    print(sample[searchTuple])
else:
    print("name not found")
```

```
enter first namesohaib
enter last nameali
0246585468445
```

Lab Task 1:

Create two lists based on the user values. Merge both the lists and display in sorted order.

```
list1 = []
list2 = []

n1 = int(input("Enter the number of elements in the first list: "))
for i in range(n1):
    val = int(input(f"Enter element {i+1} for list 1: "))
    list1.append(val)

n2 = int(input("Enter the number of elements in the second list: "))
for i in range(n2):
    val = int(input(f"Enter element {i+1} for list 2: "))
    list2.append(val)
merged_list = list1 + list2
merged_list.sort()

print("Merged and sorted list:", merged_list)
```

```
Enter the number of elements in the first list: 2
Enter element 1 for list 1: 5
Enter element 2 for list 1: 2
Enter the number of elements in the second list: 2
Enter element 1 for list 2: 8
Enter element 2 for list 2: 3
Merged and sorted list: [2, 3, 5, 8]
```

Lab Task 2:

Repeat the above activity to find the smallest and largest element of the list.

```
smallest = min(merged_list)
largest = max(merged_list)

print("Smallest element:", smallest)
print("Largest element:", largest)
```

```
Smallest element: 2
Largest element: 8
```

Lab Task 3:

The derivate of a function f(x) is a measurement of how quickly the function f…

```
import math

x_values = [x * 0.001 for x in range(-1000, 1001)]
h = 0.001
for x in x_values:
    derivative = (math.sin(x + h) - math.sin(x)) / h
    print(f"x = {x}, sin(x) = {math.sin(x)}, derivative = {derivative}")
```

```
x = -1.0, sin(x) = -0.8414709848078965, derivative = 0.5407229512750522
x = -0.999, sin(x) = -0.8409302618566215, derivative = 0.5415638814669776
x = -0.998, sin(x) = -0.8403886979751545, derivative = 0.5424042700947762
x = -0.997, sin(x) = -0.8398462937050597, derivative = 0.5432441163185642
x = -0.996, sin(x) = -0.8393030495887411, derivative = 0.5440834192981248
x = -0.995, sin(x) = -0.838758966169443, derivative = 0.5449221781944624
x = -0.994, sin(x) = -0.8382140439912485, derivative = 0.5457603921686927
x = -0.993, sin(x) = -0.8376682835990799, derivative = 0.546598060382375
x = -0.992, sin(x) = -0.8371216855386975, derivative = 0.5474351819981793
x = -0.991, sin(x) = -0.8365742503566993, derivative = 0.5482717561787753
x = -0.99, sin(x) = -0.8360259786005205, derivative = 0.549107782087721
x = -0.989, sin(x) = -0.8354768708184328, derivative = 0.5499432588890185
x = -0.988, sin(x) = -0.8349269275595438, derivative = 0.5507781857468919
x = -0.987, sin(x) = -0.8343761493737969, derivative = 0.5516125618267864
x = -0.986, sin(x) = -0.8338245368119701, derivative = 0.5524463862940365
x = -0.985, sin(x) = -0.8332720904256761, derivative = 0.5532796583150867
x = -0.984, sin(x) = -0.832718810767361, derivative = 0.5541123770564926
x = -0.983, sin(x) = -0.8321646983903045, derivative = 0.5549445416854759
x = -0.982, sin(x) = -0.831609753848619, derivative = 0.5557761513700354
x = -0.981, sin(x) = -0.831053977697249, derivative = 0.5566072052785032
x = -0.98, sin(x) = -0.8304973704919705, derivative = 0.5574377025797661
```

Lab Task 4:

For this exercise, you will keep track of when our friend's birthdays…

```
birthday_dict = {
    "Albert Einstein": "03/14/1879",
    "Benjamin Franklin": "01/17/1706",
    "Ada Lovelace": "12/10/1815"
}

print("Welcome to the birthday dictionary. We know the birthdays of:")
for name in birthday_dict.keys():
    print(name)

name = input("Who's birthday do you want to look up? ")

if name in birthday_dict:
    print(f"{name}'s birthday is {birthday_dict[name]}.")
else:
    print(f"{name} not found in the dictionary.")
```

```
Welcome to the birthday dictionary. We know the birthdays of:
Albert Einstein
Benjamin Franklin
Ada Lovelace
Who's birthday do you want to look up? Benjamin Franklin
Benjamin Franklin's birthday is 01/17/1706.
```

Lab Task 5:

Write a Python program to create a new dictionary by extracting the mentioned keys from the below dictionary….

```
sample_dict = {
    "name": "Kelly",
    "age": 25,
    "salary": 8000,
    "city": "New York"
}

keys = ["name", "salary"]

new_dict = {key: sample_dict[key] for key in keys}

print("Extracted Dictionary:", new_dict)
```

```
Extracted Dictionary: {'name': 'Kelly', 'salary': 8000}
```